ZZ-B-530C
December 18, 1974
SUPERSEDING
Fed. Spec. ZZ-B-530B
May 26, 1966

FEDERAL SPECIFICATION

BOOTS, KNEE AND HIP, (RUBBER)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for heavy-duty rubber boots.

1.2 Classification. The boots covered by this specification shall be of the following types and sizes as specified (see 6.2):

1.2.1 Type.

I - Knee length.
II - Hip length.

1.2.2 Sizes. Boots shall be furnished in whole sizes 5 through 15.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

TT-E-529	-	Enamel, Alkyd, Semi-Gloss.
UU-P-268	-	Paper, Kraft, Wrapping.
PPP-P-636	-	Boxes, Shipping, Fiberboard.

Federal Standards:

FED-STD-123	-	Marking for Domestic Shipment (Civil Agencies).
FED-STD-191	-	Textile Test Methods.
FED-STD-601	-	Rubber: Sampling and Testing.

FSC 8430

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers, at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Ft. Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-B-17757 - Boxes, Fiberboard, Corrugated (Modular Sizes).

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-284 - Visual Inspection Guide for Rubber Footwear.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

U.S. Postal Service Manual

(Copies of the manual may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

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American Society for Testing and Materials (ASTM) Standards

D 377-69 - Small Amounts of Copper and Manganese in Textiles.
D 1630-61 - Abrasion Resistance of Rubber Compounds for Soles and Heels.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Rubber compounds. All rubber compounds used in the manufacture of boots shall be made from natural or synthetic rubber or a mixture of both.

3.1.1.1 Upper, outsole, and heel compounds. The compounds for the upper, outsole, and heel of the boot shall meet the physical requirements specified in table I when tested as specified in 4.2.1. The compound for the outsole and heel shall be nonmarking when tested as specified in 4.2.1.

TABLE I. Requirements for rubber compounds

Characteristics	Upper compound	Outsole and heel compound
Tensile strength, minimum:		
Initial, p.s.i.	1500	---
After accelerated aging, percent of initial strength	75	---

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TABLE I. Requirements for rubber compounds (cont'd)

Characteristics	Upper compound	Outsole and heel compound
Elongation, minimum:		
Initial, percent	400	---
After accelerated aging, percent of initial elongation	75	---
Abrasive index, minimum:		
Initial	---	35
After accelerated aging	---	35
Hardness before and after aging	---	60 ± 10

3.1.2 Cotton cloth. The base cloth used in the boots shall be dyed or undyed cotton cloth and shall conform to the requirements specified in table II. All base cloths to be coated shall contain not more than 0.003 percent copper or 0.002 percent manganese, and shall have a pH of 5.5 to 8.5. Colored base cloths which are exposed in the finished boot shall show "good" fastness or better to crocking (Munsell value not less than 8.5).

3.1.3 Buckle (for type II boots). The boot strap buckle shall be of the slide type. The buckle shall be brass, aluminum, or steel. The strap slot shall be not less than 5/8 inch wide, and the stay slot shall be not less than 7/16 inch wide.

3.1.3.1 Color and finish. Brass buckles shall be nickel plated; steel buckles shall be either nickel plated or have a black baked enamel finish; aluminum buckles shall be anodized with a coat of black enamel. The enamel shall conform to class B of TT-E-529. The coating shall be a continuous dry film without objectionable orange peel, wrinkles, drops, streaks, or uncovered areas.

3.1.4 Cord for stay reinforcement. The cord used to reinforce the buckle stay shall be a twine, having a breaking strength of 12 pounds minimum when tested as specified in 4.2.1.

3.2 Design. The boots shall be made over the manufacturer's regular boot lasts.

3.2.1 Type I boot. Type I boot shall be a knee-length vulcanized-rubber boot as illustrated in figure 1.

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3.2.2 Type II boot. Type II boot shall be a hip-length vulcanized-rubber boot with an adjustable top strap as illustrated in figure 1.

3.3 Construction.

3.3.1 Frictioned components. The term "frictioned" when used herein shall mean that the applicable cloth shall be treated with friction compound by a frictioning process. Frictioned components shall conform to the details of construction specified in table II.

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TABLE II. Requirements of components using cotton cloth

Component	Weight of cotton cloth (ounces, min.) Woven	Knitted	Frictioned Both sides	One side	Calendar coated	Spread and calender coated	Coated	Overall thickness (inch) minimum	Additional construction details
Leg form	3.2	-	X	-	-	-	-	-	-
Leg lining	6.9	-	-	X	On frictioned side with lining upper compound	-	-	0.030	-
First option	6.9	-	-	-	-	On one side only with friction compound	-	0.030	-
Ankle reinforcement	3.5	-	-	X	On the other side with lining upper compound	-	-	0.020	-
First option	3.5	-	X	-	-	-	-	-	Cloth ankle shall be joined with an upper compound ankle having a minimum thickness of 0.015 inch.

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TABLE II. Requirements of components using cotton cloth (cont'd)

Component	Weight of cotton cloth (ounces, min.) Woven	Knitted	Frictioned Both sides	One side	Calendar coated	Spread and calender coated	Coated	Overall thickness (inch) minimum	Additional construction details
Ankle reinforcement (cont'd)									
Second option	-	-	-	-	-	-	-	-	If the upper compound ankle is not less than 0.030 inch thick, the cloth ankle may be omitted.
Toe lining	5.4	-	-	X	-	-	-	0.025	-
First option	5.4	-	-	X	-	On the same side with lining upper compound	-	0.025	-
Second option	-	9.0	-	-	On one side with friction compound	-	-	0.040	-
Third option	-	6.9	-	-	On one side with lining upper compound	-	-	0.030	-
Fourth option	5.4	-	-	X	-	-	On the same side with friction compound	0.030	-

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TABLE II. Requirements of components using cotton cloth (cont'd)

Component	Weight of cotton cloth (ounces, min.) Woven	Knitted	Frictioned Both sides	One side	Calender coated	Spread and calender coated	Coated	Overall thickness (inch) minimum	Additional construction details
Inner vamp	-	-	-	-	-	-	-	-	Shall be made from upper compound not less than 0.020 inch thick.
First option	5.0	-	X	-	-	-	-	0.018	-
Second option	5.0	-	X	-	-	-	On one side with friction compound	0.025	-
Third option	-	-	-	-	-	-	-	-	Not required if the toe lining is made from 5.4 ounce cloth that has been frictioned and coated with friction compound to an overall thickness of 0.030 inch. Not required if the toe lining is made from 6.9 ounce cloth that has been calender coated to an overall thickness of 0.030 inch.

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TABLE II. Requirements of components using cotton cloth (cont'd)

Component	Weight of cotton cloth (ounces, min.) Woven	Knitted	Frictioned Both sides	Frictioned One side	Calender coated	Spread and calender coated	Coated	Overall thickness (inch) minimum	Additional construction details
Vamp form	5.0	-	X	-	-	-	-	-	One side shall be covered with upper compound vamp (see table III).
Friction, back strip or backstay	4.8	-	X	-	-	-	-	-	The use of this component is optional.
Counter form	5.0	-	X	-	-	-	-	-	One side shall be covered with upper compound (see table III).
Insole	5.0	-	-	-	-	-	On one side with stiffening upper compound	0.100	-
Filler and binder sole	2.09	-	-	X	-	-	On the other side with stiffening upper compound	0.215 when combined with insole	The filler sole shall be made from stiffening upper compound and combined with the binder sole.

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TABLE II. Requirements of components using cotton cloth (cont'd)

Component	Weight of cotton cloth (ounces, min.) Woven	Knitted	Frictioned Both sides	One side	Calender coated	Spread and calender coated	Coated	Overall thickness (inch) minimum	Additional construction details
Filler and binder sole (cont'd)									
First option	2.3	-	X	-	-	-	On one side with stiffening upper compound	0.215 when combined with insole	-
Second option	-	-	-	-	-	-	-	0.215 when combined with insole	The filler and binder sole shall be combined and made of stiffening upper compound.
Stiffening counter	3.5	-	-	X	-	-	On the other side with stiffening upper compound	0.045	-
Top lining	6.0	-	-	X	-	-	-	-	-
First option	-	7.2	-	-	-	-	With lining upper compound	-	-

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TABLE II. Requirements of components using cotton cloth (cont'd)

Component	Weight of cotton cloth (ounces, min.) Woven	Knitted	Frictioned Both sides	One side	Calender coated	Spread and calender coated	Coated	Overall thickness (inch) minimum	Additional construction details
Top buckle stay (type II)	5.0	-	X	-	-	-	On one side with upper compound	0.025	The stay shall be folded and shall include two reinforcement cords of twine, placed inside the stay, one cord at each edge. The folded stay shall be approximately 7/16 inch wide.

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3.3.2 Upper compound components. The upper compound components shall conform to the details of construction specified in table III and shall be made of the rubber compound specified.

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TABLE III. Upper compound components

Component	Type of rubber compound	Applicable boot type	Thickness (inch) (minimum)	Additional construction details
Leg cover	Upper	I and II	0.025	-
Back strip	Upper	I and II	0.020	If a friction back strip is used, the edges of the upper compound strip shall overlap those of the friction strip and be securely rolled to the upper compound leg cover.
Combined backstay and counter (if used)	Upper	I and II	0.025 (combined)	-
Vamp	Upper	I and II	0.030	-
Counter	Upper	I and II	0.025	-
Top reinforcement	Upper	I and II	0.015	The top reinforcement shall be adhered to the top lining on the inside of the top either on the inner facing of the top lining, or between the top lining and the top cover. The top reinforcement shall be not less than 3 inches wide.
Binding	Outsole and heel	I and II	0.030	The binding shall be not less than 5/16 inch wide and shall be securely bonded to the outside top edge of the boot. A black enameled rivet and cap or an eyelet may be inserted at the binding or just below the binding at the back of the boot.
Foxing	Outsole and heel	I and II	0.030	The foxing shall be applied before the outsole and shall extend completely around the bottom edges of the boot.
Outsole	Outsole and heel	I and II	0.285 at the ball and 0.100	The outsole shall be cleated or heavily knurled and secured to the boot after foxing.

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TABLE III. Upper compound components (cont'd)

Component	Type of rubber compound	Applicable boot type	Thickness (inch) (minimum)	Additional construction details
Heels	Outsole and heel	I and II	9/16 at thinnest part	Heels shall carry the manufacturer's standard tread design.
Top	Upper	II	0.025	Where a tapered roll is used, the minimum thickness shall be 0.020 inch.
Knee patch	Upper	II	0.020	The knee patch shall be not less than 60 square inches in area and shall be so placed to prevent chafing of the top where the legs rub together.

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3.3.3 Top strap. Each type II boot shall be furnished with an adjustable strap made with a loop at one end, through which the belt of the wearer may be slipped. The strap shall be not less than 9-1/2 inches long by 5/8 inch wide. It shall be either a double cotton cloth frictioned on both sides and coated with upper compound, or a rubber strap not less than 0.090 inch thick. The loop at the top of the strap shall be large enough to permit easy passage and secure holding over a 2-inch belt.

3.4 Measurements.

3.4.1 Height. The height for type I boots inside at the back and front shall be not less than 14-1/2 inches. The extreme height for type II boots inside shall be not less than 34 inches and the crotch height of the boots inside shall be not less than 29 inches.

3.4.2 Girth of last. The circumference at the calf (largest circumference between knee and ankle) of the boot last, over which the boots are made, shall be not less than:

16 inches for sizes 5, 6, 7.
16-1/2 inches for sizes 8 and 9.
17 inches for sizes 10 and 11.
18 inches for sizes 12, 13, 14, and 15.

3.4.3 Girth (at crotch). For type II boots, the girth measured around the inside of the boot at the crotch shall be not less than 23 inches.

3.5 Color and finish. The finished boots shall be black and shall have a natural heater or dull-lacquer finish.

3.6 Marking.

3.6.1 Trademark, name, or label. The manufacturer's name or trademark shall be embossed on the outsole in the shank or a manufacturer's label shall be used. If a manufacturer's label is used, it shall be not more than 1-1/2 inches in diameter and not more than one label shall be used on each boot. The label may be placed at any suitable position on the outside upper of the boot.

3.6.2 Size. The size shall be legibly incised on the outsole in the shank near the breast of the heel. Size markings shall be accomplished in such a manner that the outsole is not weakened or damaged. Figures used for size identification shall be not less than 1/2 inch in height. As an alternate method, the size may be legibly incised on a label and the label permanently vulcanized to the boot, either on the outsole edge near the heel, or at the top of the boot at the back.

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3.6.3 Contractual data. The contract number and size of the boot shall be legibly stamped with indelible ink in a contrasting color at the top of the lining of the boot. Letters shall be not less than 1/4 inch in height.

3.7 Vulcanization. Boots shall be vulcanized under pressure to insure a watertight compact unit.

3.8 Leakage. Boots shall show no evidence of leakage when tested as described in 4.2.3.2.

3.9 Replacement of defective components. During the cutting and manufacturing process, components having material defects or damages that are classified as defects in 4.2.3.1.1 shall be removed from production and replaced with non-defective and properly matched components.

3.10 Workmanship. The finished boots shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.1.1 Testing of components and materials. In addition to the quality assurance provisions of applicable referenced specifications and standards, testing shall be performed on components and materials listed in table IV for the characteristics shown. The sample unit for each component or material shall be as follows:

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(a) Base cloth (natural) - One square foot.
(b) Base cloth (dyed) - Two square feet.
(c) Upper compound - Two pieces 6 by 6 inches by 0.070 ± 0.010 inch.
(d) Outsole and heel compound - Two pieces 6 by 6 inches by 0.250 ± 0.025 inch.
(e) Cord or equivalent twine - Ten yards.

The sample size for all components or materials except rubber compounds shall be as specified below and the lot shall be unacceptable if any unit fails to meet any requirement specified.

Lot size	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and more	5

For each type of rubber compound, a lot shall consist of the amount of rubber compound that is produced or compounded in one day. The composite sample of rubber shall be vulcanized in the same manner employed in the manufacture of the boots. The sample size shall be 1 and the lot shall be unacceptable if any unit fails to meet any requirement specified. The physical values specified apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods.

TABLE IV. Instructions for testing materials and components

Component or material and lot size expressed in units of	Characteristic	Requirement reference	Test method
Cotton cloth, each type and weight (1 yard full width)	Material identification 1/	Table II and 3.3.3	1200 2/
	Weight 1/	Table II	5041 2/
	Colorfastness (for dyed cloths only):		
	- to perspiration	3.1.2	5680 2/
	- to crocking	3.1.2	5651 2/
	Copper manganese content 1/	3.1.2	D 377-69 3/ (photometric method)
	Acidity (pH) 1/	3.1.2	2811 2/

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TABLE IV. Instructions for testing materials and components (cont'd)

Component or material and lot size expressed in units of	Characteristic	Requirement reference	Test method
Cord (1 pound)	Breaking strength	3.1.4	4102 2/
Upper compound	Tensile strength:		
	- initial	Table I	4111 4/
	- after accelerated aging	Table I	7001, 7221, and 4111 4/ 5/
	Elongation:		
	- initial	Table I	4121 4/
	- after accelerated aging	Table I	7001, 7221, and 4121 4/
Outsole and heel compound	Abrasive index:		
	- initial	Table I	D 1630-61 3/
	- after accelerated aging	Table I	D 1630-61 3/ 6/
	Hardness	Table I	3021 4/
	Nonmarking	3.1.1.1	4.3.1

1/ The supplier shall submit a certificate of compliance for these characteristics (see 4.2.1.1.1). In case of dispute, the method specified in the table shall be used.

2/ FED-STD-191.

3/ ASTM Standards.

4/ FED-STD-601.

5/ Aging shall be conducted at 212° ± 4°F for 24 hours.

6/ Aging shall be conducted at 212° ± 5°F for 24 hours.

4.2.1.1.1 Certificate of compliance. A certificate of compliance shall be furnished with each shipment or lot stating the following:

(a) That the boots were made with natural or synthetic rubber or a mixture of both.
(b) That the metal in the buckle (when applicable) is as specified (see 3.1.3).

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(c) The type and weight of each cotton cloth and where each was used (see tables II and IV).
(d) That all base cloths used meet the specified copper and manganese content requirements (see 3.1.2 and table IV).

4.2.2 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing process to determine whether operations or assemblies are accomplished as specified. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated nonconformance.

4.2.3 Inspection of the end items.

4.2.3.1 Examination of the end item. The defects found during the examination shall be classified in accordance with 4.2.3.1.1 and 4.2.3.1.2. The inspection level and the acceptable quality level (AQL) shall be as specified in 4.2.3.1.3. The lot size shall be expressed in units of one boot.

4.2.3.1.1 Visual examination. The boots shall be examined for the defects listed below and the applicable defects established in MIL-STD-284. The sample unit shall be one completely fabricated boot and the selection shall be in pairs. For pairing examination, the pair shall be examined together. Each defect found during the examination for pairing shall be scored as a single defect.

Defect	Classification Major	Minor
More than 1/4 inch variation in height of pair		X
Color not black		X
Design not as specified		X
Poor or uneven lasting, seriously affecting the serviceability or appearance	X	
Poor or uneven lasting, affecting serviceability or appearance but not seriously		X
Air pocket or channel under 1/4 square inch in area in any seam		X
Air pocket or channel 1/4 square inch or more in area in any seam	X	
Marking, (contractural, size, etc.) missing, incomplete, incorrect, not applied in the specified manner, misplaced, or illegible		X

4.2.3.1.2 Dimensional examination. The boots shall be examined for defects in dimensions. Any dimension not within the established tolerance shall be classified as a defect. The sample unit shall be one boot.

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4.2.3.1.3 Inspection level and AQL. The inspection levels and the AQLs expressed in defects per 100 units shall be as follows:

Examination	Inspection level	AQL Major	AQL Totals
4.2.3.1.1	II	2.5	6.5
4.2.3.1.2	S-3	-	4.0

4.2.3.2 Testing of the end item. The finished boots shall be tested in accordance with the test method specified in 4.3.2 for conformance to the leakage requirement in 3.8. The sample unit shall be one boot and the requirement shall be applicable to the sample unit. The inspection level shall be S-3 and the AQL shall be 1.5 defects per 100 units. One determination shall be made for each sample unit and the results shall be reported as pass or fail.

4.2.4 Examination of preparation for delivery requirements. An examination shall be made to determine that the packaging, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery except that it need not be closed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.
Materials	Any component missing, damaged or not as specified.
Workmanship	Inadequate application of components such as: incomplete closure of container flaps, loose strapping, inadequate stapling, or improper taping. Open and noncontinuous heat sealed seams and closure of plastic bag. Incorrectly fabricated bag. Bulged or distorted container.
Content	Number of pairs of boots per container is more or less than required.

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4.3 Test methods.

4.3.1 Nonmarking test. One corner of the stock shall be drawn firmly across a sheet of white bond paper. If the mark made can be easily erased by rubbing with the fingers, the compound being tested shall be determined as being a non-marking compound. One determination per sample unit shall be made.

4.3.2 Leakage test. Boots shall be tested for leakage by inserting a hose or pipe, connected to a source of compressed air at 1 pound per square inch pressure, well down into the boot. The top of the boot shall be closed in a manner to prevent the escape of air and the boot shall then be immersed in a tank of water to within 3 inches of the top of the boot. Evidence of leakage will be indicated by bubbles of air that form and rise to the surface of the water.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Each mated pair of boots shall be tied together with cotton tape or twine. Alternately each mated pair of boots shall be placed in a snug-fitting clear plastic film bag having a thickness of 0.0015 (± 20 percent tolerance). The bag shall be formed with heat sealed seams that are straight, continuous and parallel to each other and the formed edges of the bag. The bag may be fabricated from plastic film tubing or sheeting. The open end of the bag shall extend a minimum of 2-inches beyond the packaged length of the boots.

5.1.2 Level C (commercial packaging). Boots shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The package and the quantity per package shall be the same as that normally needed by the supplier for retail distribution.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Seven pairs of type I (knee) boots or six pair of type II (hip) boots of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled, closed, and reinforced conforming to class weather-resistant, grade V2s, size 3A of MIL-B-17757. The fiberboard for the box liner shall conform to type CF, class weather-resistant, variety DW, grade V15c of MIL-B-17757. Prior to placing each pair of hip boots in the container, the tops shall be folded over so that the packing can be accomplished as specified. Level A packages shall be packed within a shipping container so that the first pair of boots has the sole of one boot against one end panel and the side of the sole of the second boot resting on the ankle of the first boot. The remaining pairs of boots shall then be placed in the container in the same manner with every other pair reversed. A sheet of 30-pound minimum basis weight kraft paper conforming to grade B of UU-P-268 shall be used to completely separate each boot from another boot and to completely cover the top and

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bottom of each pack. When boots are packaged in plastic bags the kraft paper separators shall not be required. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes of one type may be packed within the same container. Containers of less than full pack multiples shall have all void spaces filled with folded or scored corrugated fiberboard pads to immobilize the contents.

5.2.2 Level B.

5.2.2.1 Civil agencies. Seven pairs of type I (knee) boots or six pairs of type II (hip) boots of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style OSC-L, type CF or SF, class domestic, variety SW, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to class domestic, variety SW, grade 275 of PPP-B-636. Prior to placing each pair of hip boots in the container, the tops shall be folded over so that the packing can be accomplished as specified. Level A packages shall be packed within a shipping container so that the first pair of boots has the sole of one boot against one end panel and the side of the sole of the second boot resting on the ankle of the first boot. The remaining pairs of boots shall then be placed in the container in the same manner with every other pair reversed. A sheet of 30-pound minimum basis weight kraft paper conforming to grade B of UU-P-268, shall be used to completely separate each boot from one another and to completely cover the top and bottom of each pack. Inside dimensions of each shipping container shall be 23-1/2 inches in length, 15 inches in width, and 15 inches in depth. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636. When boots are packaged in plastic bags the kraft paper separators shall not be required. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes of one type may be packed within the same container. Containers of less than full pack multiples shall have all void spaces filled with folded or scored corrugated fiberboard pads to immobilize the contents.

5.2.2.2 Military requirements. Seven pairs of type I (knee) boots or six pairs of type II (hip) boots of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled and closed conforming to class domestic, grade 275, size 3A of MIL-B-17757. The fiberboard for the box liner shall conform to type CF, class domestic, variety DW, grade 275 of MIL-B-17757. Prior to placing each pair of hip boots in the container, the tops shall be folded over so that the packing can be accomplished as specified. Level A packages shall be packed within a shipping container so that the first pair of boots has the sole of one boot against one end panel and the side of the sole of the second boot resting on the ankle of the first boot. The remaining pairs of boots shall then be placed in the container in the same manner with every other pair reversed. A sheet of 30-pound minimum basis weight kraft paper conforming to grade B of UU-P-268 shall be used to

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completely separate each boot from one another and to completely cover the top and bottom of each pack. When boots are packaged in plastic bags the kraft paper separators shall not be required. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes of one type may be packed within the same container. Containers of less than full pack multiples shall have all void spaces filled with folded or scored corrugated fiberboard pads to immobilize the contents.

5.2.2.2.1 When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with MIL-B-17757 and closed in accordance with appendix of the container specification.

5.2.3 Level C (commercial packing). Boots, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the supplier for retail distribution. Containers shall comply with the US Postal Service Manual, Uniform Freight Classification Rules, or National Motor Freight Classification Rules, as applicable.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with FED-STD-123.

5.3.2 Military requirements. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

5.3.3 Labels, mixed sizes. Each shipping container packed with mixed sizes shall have securely attached to the end and side a white paper label 5 by 4 inches with the words "MIXED SIZES" plainly stamped or printed thereon and under these words shall be legibly stamped or printed the correct quantity and sizes contained therein.

6. NOTES

6.1 Intended use. The boots are ordinary service type and are resistant to water only.

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FIG. 1-BOOTS, KNEE AND HIP (RUBBER)

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) The type and size of boots required (see 1.2.1 and 1.2.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.2.1).

MILITARY CUSTODIANS:

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Air Force - 82

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Navy - YD, MC, CG

Preparing activity:

Army - GL

Civil Agency Coordinating Activity:

GSA - FSS

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DOCUMENT IDENTIFIER AND TITLE

ZZ-B-530 - Boots, Knee and Hip (Rubber)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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